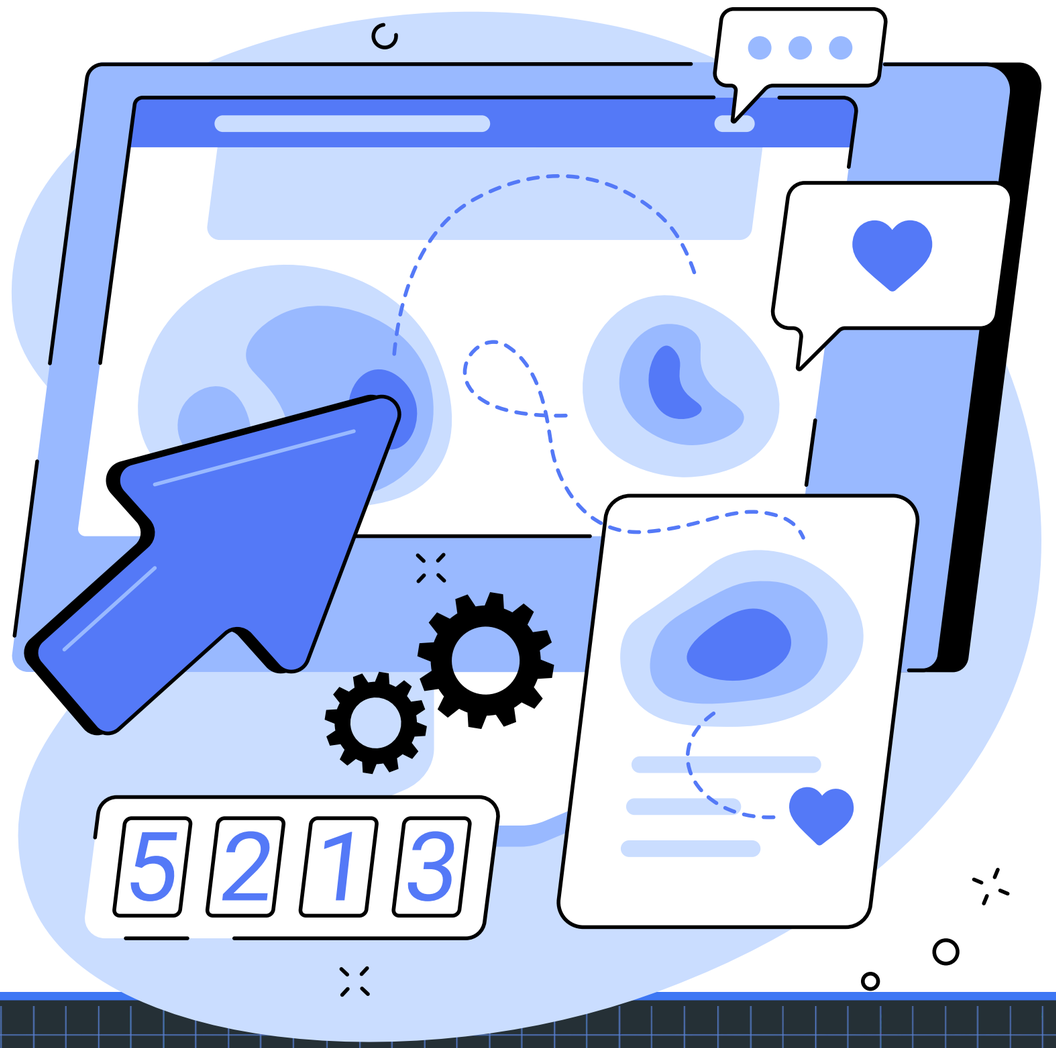
DECODING TECH HIRING TRENDS:

A KEYWORD ANALYSIS OF LINKEDIN JOB POSTINGS

Phuong Anh Phi & Charlene Vu



Why This Topic?

- The tech job market is highly competitive and rapidly evolving
- Job postings signal **what skills employers value in real time**
- Students rely on these implicit signals when choosing majors, courses, and career paths
- Research exists on **labor trends**, but less on *fine-grained, role-specific* skill keywords





Implications For Stakeholders

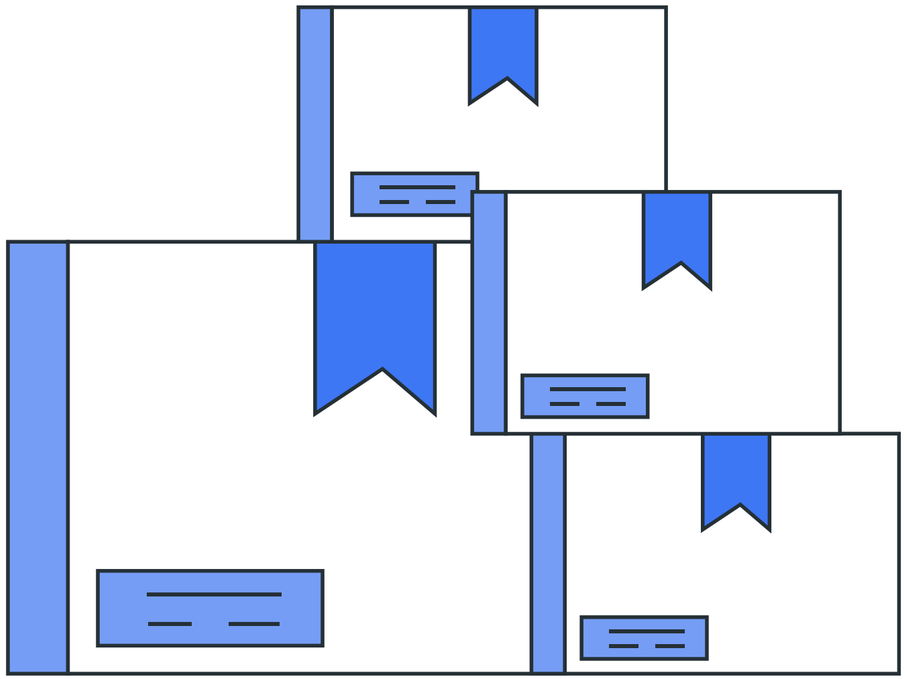
Students / Job Seekers: Identifies high-demand skills for prioritizing learning

Universities: Helps align curriculum with industry needs

Employers: Benchmark skill requirements against competitors

Researchers: Contributes to empirical labor market analytics

RESEARCH GOAL



Identify which skills and qualifications appear most frequently in tech job postings, and how they vary by role

- Which keywords are most common across CS/data roles?
- How do requirements differ between Data Analyst vs. Data Scientist vs. Software Engineer?
- Are there emerging tools or skills trending upwards?

DATA OVERVIEW

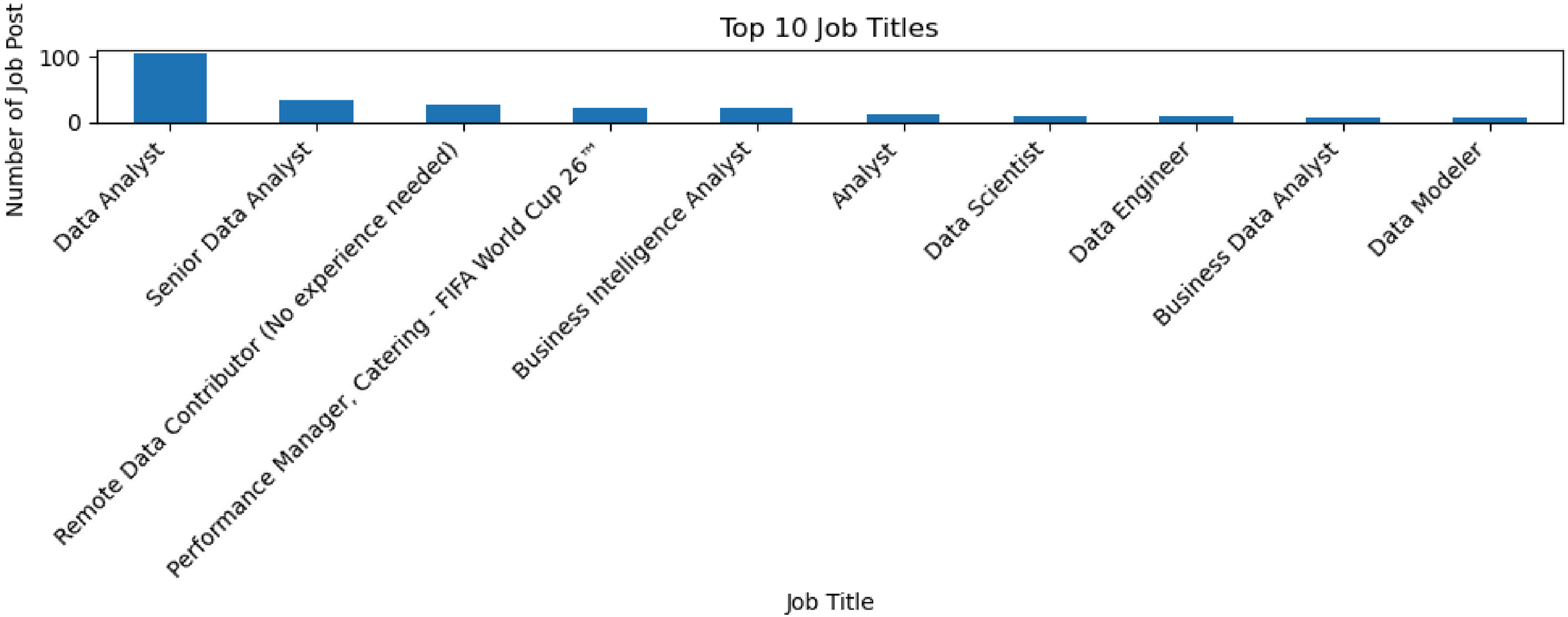
Our Dataset

Source: LinkedIn job postings
Method: Web scraping
Scope:

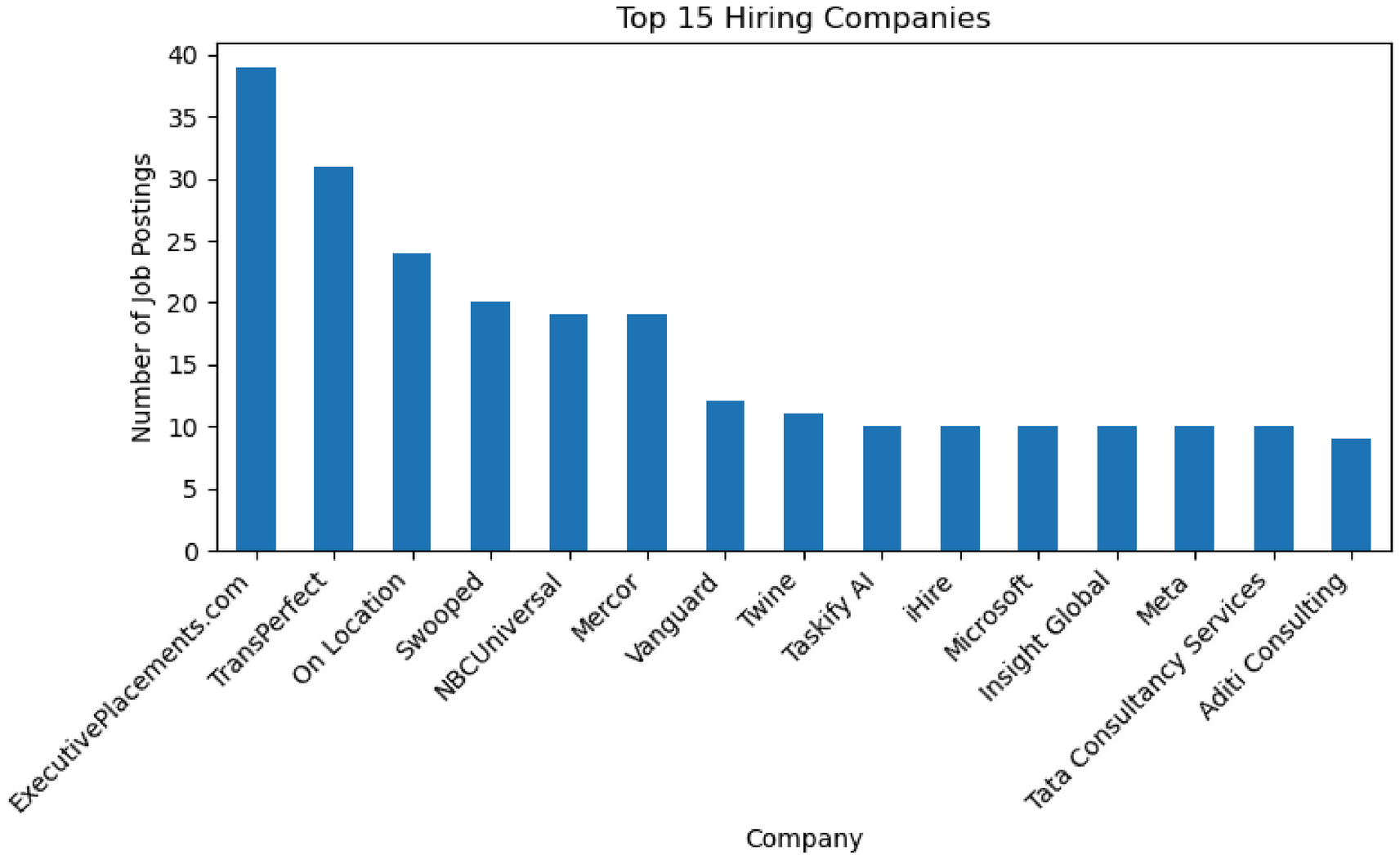
- Platform: LinkedIn
- Region: US
- Roles scraped: Data Science / Data Analyst

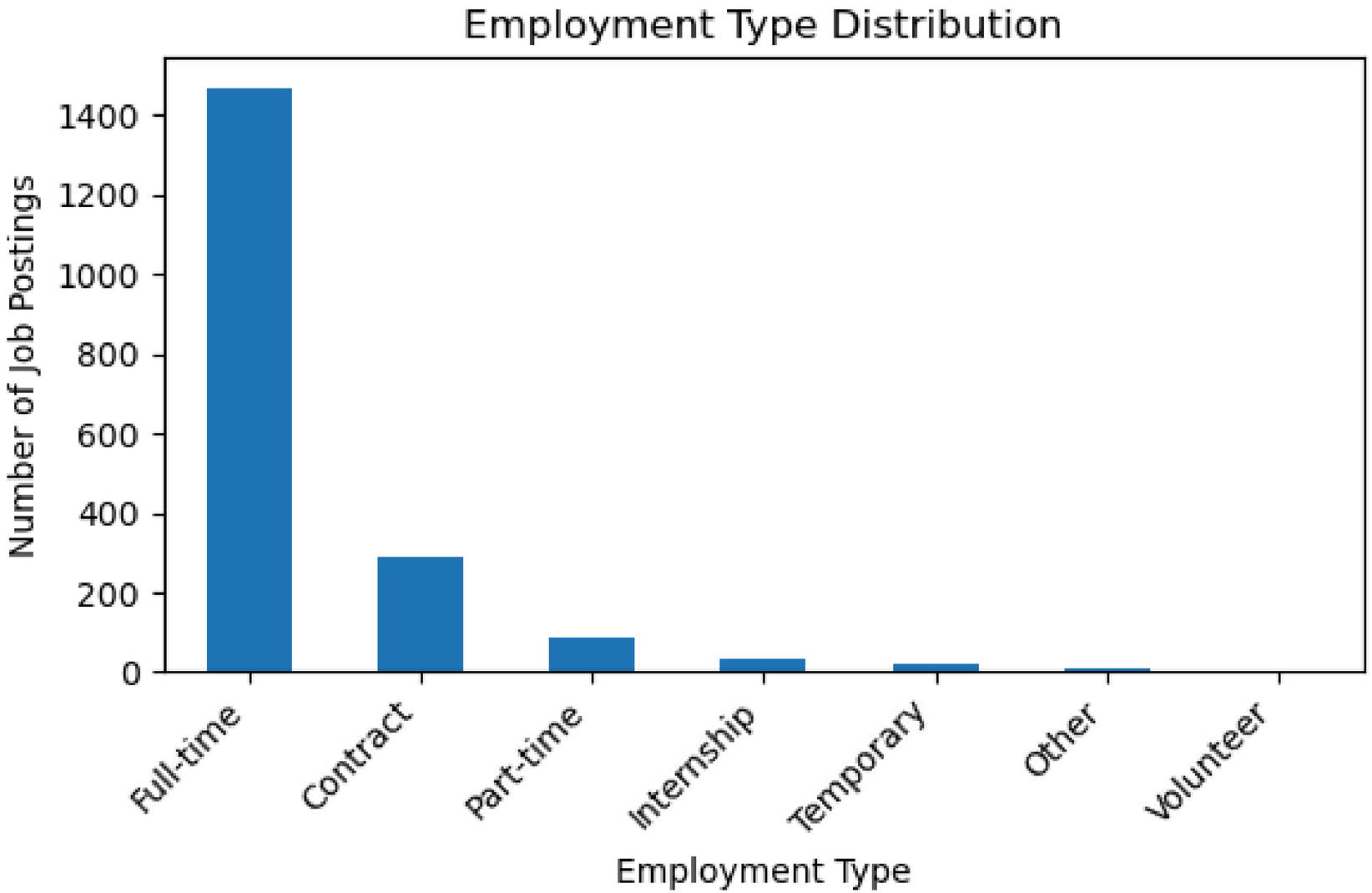
- Initially had 3778 observations and 34 variables
- Final dataset significantly refined (to 1909 observations) due to significant number of duplicates
- Removed 10 variables

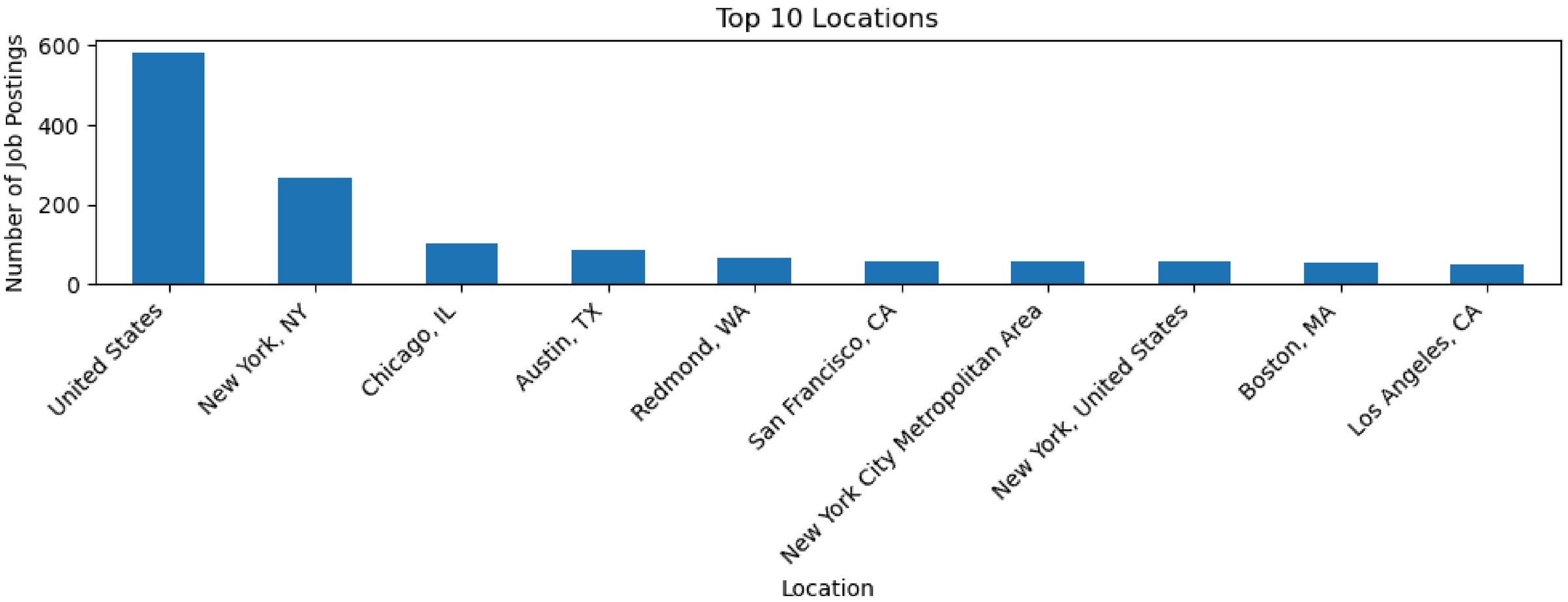


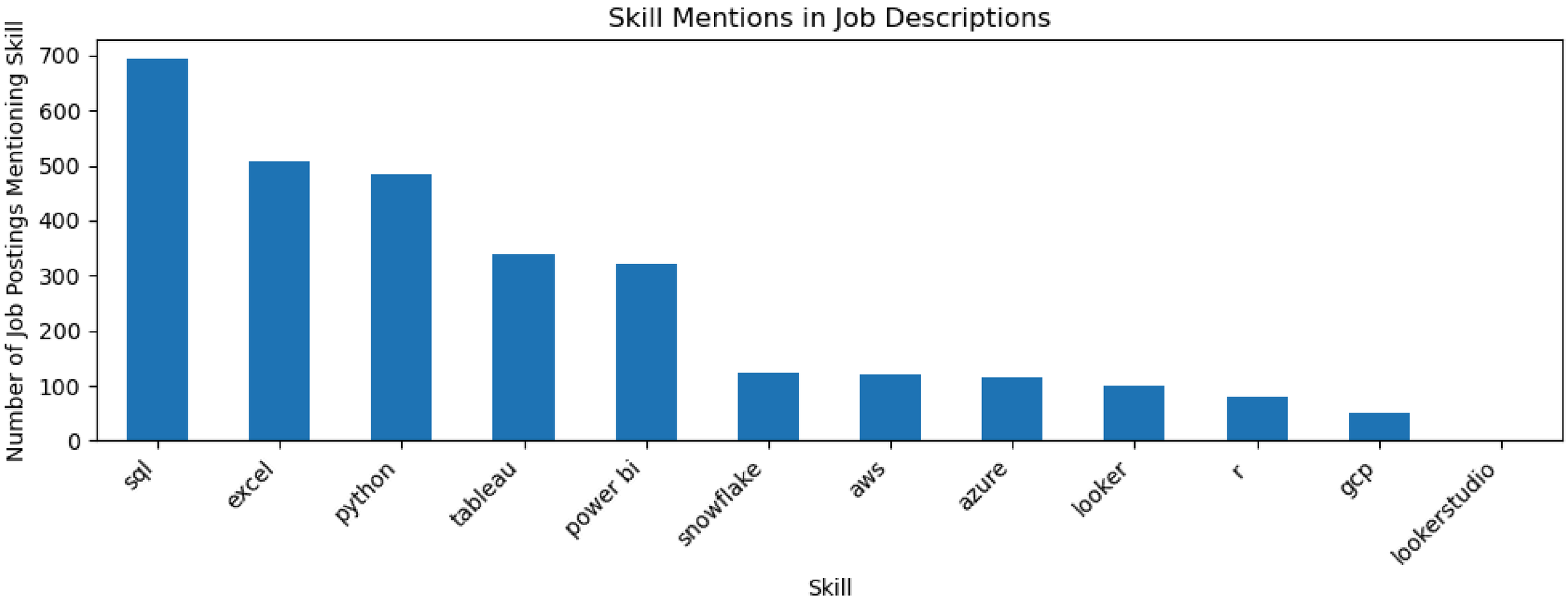

Top 10 Job Titles
Number of Job Post
100
0
Data Analyst
Senior Data Analyst
Remote Data Contributor (No experience needed)
Performance Manager, Catering - FIFA World Cup 26™
Business Intelligence Analyst
Analyst
Data Scientist
Data Engineer
Business Data Analyst
Data Modeler
Job Title

EXPLORATORY DATA ANALYSIS




Employment Type Distribution
Number of Job Postings
0
200
400
600
800
1000
1200
1400
Full-time
Contract
Part-time
Internship
Temporary
Other
Volunteer
Employment Type


Top 10 Locations
Number of Job Postings
600
400
200
0
United States
New York, NY
Chicago, IL
Austin, TX
Redmond, WA
San Francisco, CA
New York City Metropolitan Area
New York, United States
Boston, MA
Los Angeles, CA
Location

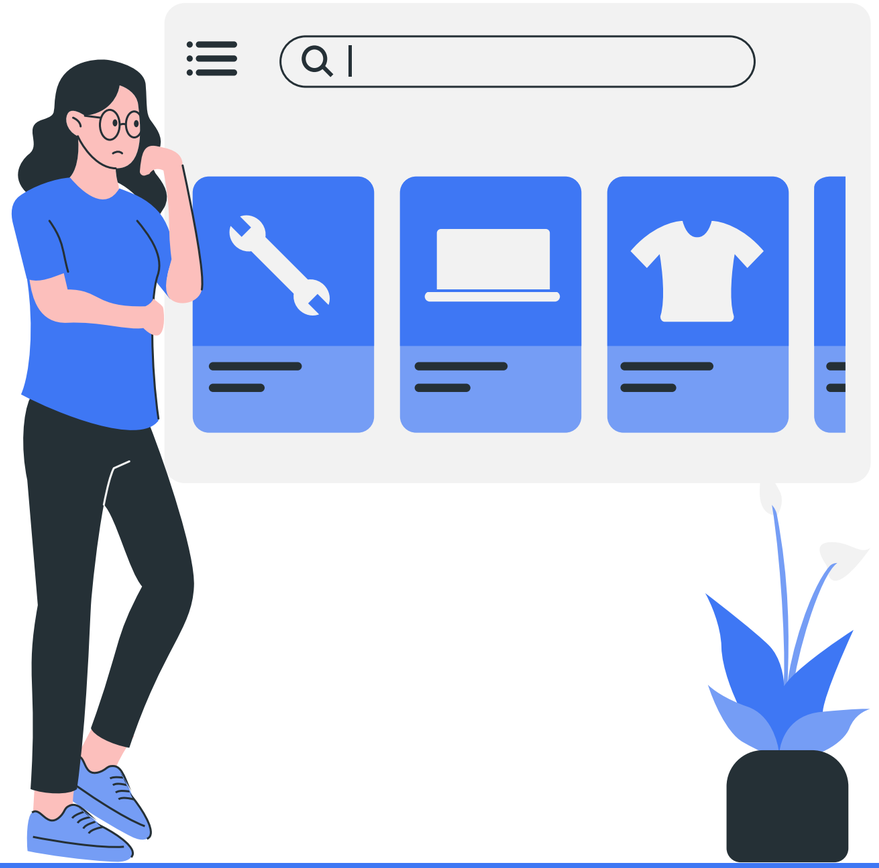

Skill Mentions in Job Descriptions
Number of Job Postings Mentioning Skill
0
100
200
300
400
500
600
700
sql
excel
python
tableau
power bi
snowflake
aws
azure
looker
r
gcp
lookerstudio
Skill

Ethical & Legal Implications

- Respecting platform terms. Used for research purpose only
- Treat data as aggregate. Avoid quoting postings or identifying employers
- Include bias analysis, acknowledge systemic hiring patterns
- Transparency in methodology: document every steps, assumptions, and limitations



FUTURE PLAN



Plan For Next Weeks

- More web scraping for a bigger dataset
- Data cleaning and text processing
- Clustering, topic modeling
- Output: word clouds, bar chart of top skills, cluster heatmap, geographic filtering

Thank You For Listening

